

# NutriBot

AI-Powered Personalized Nutrition Assistant

IBM watsonx.ai Powered Chatbot

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## OVERVIEW

# Problem Statement & Objectives

## Why NutriBot?

Bridging the gap between people and reliable, personalized nutritional guidance using the power of Generative AI.

1

### The Problem

People lack access to personalized nutrition guidance; online information is often confusing, unreliable, or requires expensive expert consultation

2

### Growing Need

Healthy meal planning demands expert knowledge, and there is a rapidly growing demand for AI-powered intelligent health assistants

3

### Our Objectives

Develop an intelligent nutrition chatbot that delivers personalized dietary recommendations and promotes healthy eating through conversational AI

## ARCHITECTURE

# System Architecture

### Frontend

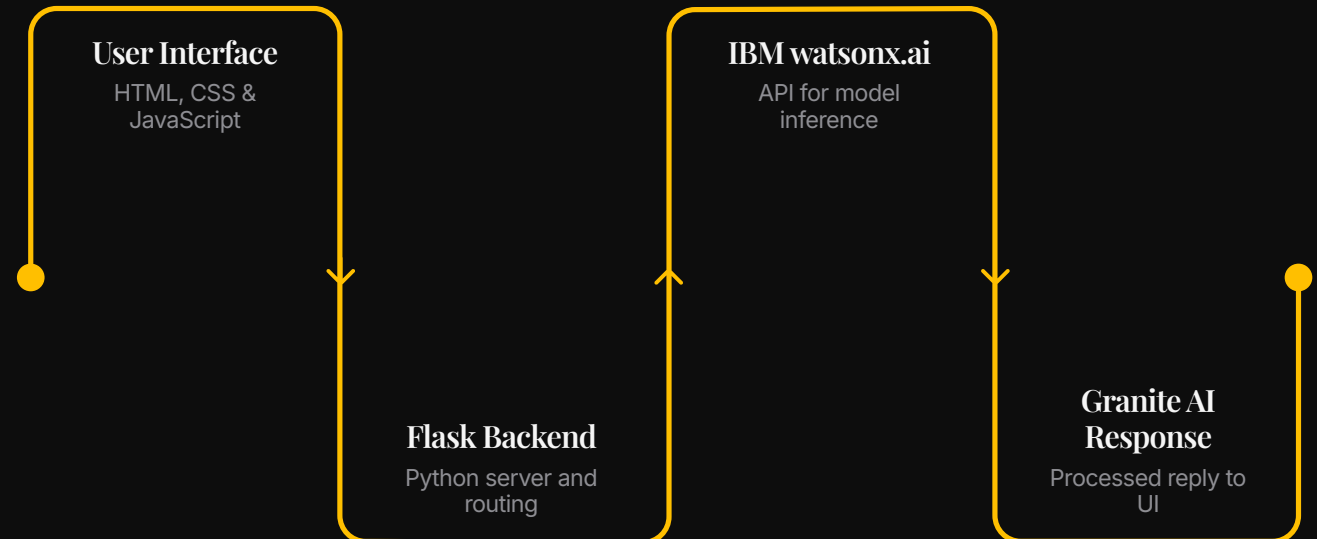
HTML, CSS & JavaScript user interface for entering queries and displaying AI responses

### Flask Backend

Python-powered Flask server processes requests and bridges the frontend with IBM watsonx.ai API

### IBM watsonx.ai

Granite Foundation Model authenticates via IBM Cloud API and generates intelligent, personalized nutrition responses



STACK

# Technology Stack

| Category      | Technologies                             |
|---------------|------------------------------------------|
| Frontend      | HTML, CSS, JavaScript                    |
| Backend       | Python, Flask                            |
| AI Platform   | IBM watsonx.ai, Granite Foundation Model |
| Configuration | Python Dotenv, REST API                  |
| Development   | Git, Virtual Environment                 |



## Web Frontend

Responsive chat interface for seamless user interaction



## AI Backend

Scalable IBM Cloud-powered intelligence engine

## WORKFLOW

# How NutriBot Works



NutriBot seamlessly connects the user's question to a powerful AI engine and returns intelligent, personalized nutrition guidance in real time.

## Key Capabilities

- **Natural Language Understanding**  
Interprets free-text nutrition questions contextually
- **Personalized Recommendations**  
Tailors dietary advice to the individual user
- **Real-time AI Responses**  
Instant answers powered by Granite Foundation Model

## FEATURES

# Features & Applications

## What NutriBot Offers

A versatile AI-powered nutrition platform serving individuals, healthcare professionals, and organizations alike.

### Key Features

1

#### AI Nutrition Assistant

Personalized meal suggestions and healthy diet recommendations via interactive chatbot

2

#### Intelligent Responses

Fast, accurate, AI-generated answers with secure IBM Cloud integration

3

#### User-Friendly Interface

Simple conversational UI accessible to all users without technical expertise

### Applications

- Personal health management
- Fitness and wellness coaching
- Hospitals and clinics
- Dietitian support tools
- Educational institutions
- Corporate wellness programs





**NutriBot** — Your Personalized AI Nutrition Coach

Chat

Dashboard

Meal Plan

BMI

Family

Profile



Mock Mode



# NutriBot

Your Personalized AI Nutrition Coach

Powered by IBM Watsonx.ai · Granite Foundation Models

Start Chatting

Calculate BMI

Dashboard

Personalized Plans

Indian Cuisine

Family Profiles

Calorie Analysis

BMI & Macros

Dark Mode



## Meal Planner

Generate AI-powered personalized meal plans

### Plan Settings

#### Plan Duration

1 Week

#### Diet Type

Balanced (All foods)

#### Primary Goal

Maintain Weight

#### Regional Cuisine Preference

Pan-Indian (Mixed)

#### Additional Preferences

e.g., no onion-garlic, quick meals under 30 min, budget friendly...

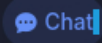
### Your Meal Plan



Configure your preferences and click  
**"Generate Meal Plan"** to get started.

The AI will generate a personalized meal plan based on your profile and preferences.





NutriBot

Ask anything about nutrition, meals, health...



👋 Hello! I'm **NutriBot**, your AI-powered nutrition coach!

I can help you with:

- 🍽️ Personalized meal plans & Indian recipes
- 📊 Calorie & nutrition analysis
- ⚖️ BMI, macros & water intake
- 👨👩👧👦 Family diet recommendations
- 🌱 Weight loss / gain strategies
- 💊 Supplements & vitamins guidance

What would you like to know today?

Now



Healthy Breakfast



Weight Loss Diet



Veggie Meal Plan



Protein Sources



Immunity Boost



Hydration Tip



### Quick Nutrition Tips

- Eat a rainbow — varied colors = varied nutrients
- Dal + rice = complete protein combination
- Drink water before meals to manage portions
- Include seasonal fruits and vegetables daily
- Haldi (turmeric) milk supports immunity

🔄 Refresh Tips



### Analyze a Food

e.g., 2 chapati with dal and sabzi...

📊 Analyze Nutrition



### Your Profile

[Edit](#)

[Set up profile](#)

GROWTH

# Advantages & Future Enhancements

## Current Advantages

Available 24x7 with a user-friendly interface; delivers accurate AI-generated responses on a scalable, cloud-integrated architecture that improves nutrition awareness

## Near-Term Enhancements

Voice-enabled chatbot, multilingual support, image-based food recognition, and a calorie tracking dashboard

## Long-Term Vision

Mobile application, wearable device integration, and personalized health analytics for comprehensive wellness management

# Conclusion & Thank You

## Transforming Nutrition Guidance with Generative AI

NutriBot demonstrates how IBM watsonx.ai and the Granite Foundation Model can power an intelligent healthcare chatbot that delivers personalized nutritional guidance through natural language conversations.

Generative AI in healthcare makes nutrition advice more accessible, interactive, and efficient for everyone.

### Project Impact

Practical application of Generative AI in healthcare, bridging the gap between users and expert nutritional knowledge

### Technology Showcase

End-to-end integration of IBM watsonx.ai, Granite Foundation Model, Flask, and a responsive web frontend

### Thank You

Questions & Discussion — Presented by Ansley Golda B, B.E CSE, J.N.N Institute of Engineering

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# Bhuvanesh P

Has successfully satisfied the requirements for:

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